

Biomes and Aquatic Ecosystems

KEY QUESTIONS

- What abiotic and biotic factors characterize a biome?
- What factors shape aquatic ecosystems?
- What are the major categories of freshwater ecosystems?
- Why are estuaries so important?

HS-LS2-2: Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales.

VOCABULARY

biome
canopy
understory
humus
taiga
permafrost
photic zone
aphotic zone
plankton
wetland
estuary

READING TOOL

As you read, note the similarities and differences between the different land biomes and aquatic ecosystems. Take notes in your **Biology Foundations Workbook**.

VIDEO

Watch this video to learn about the tundra biome and its inhabitants.



Golden lion tamarin, adult and baby

The factors that create regional climates establish conditions that govern which plants, animals, and other organisms can survive in particular geographic areas. These areas are described in this lesson.

Life on Land: Natural Biomes

Ecologists typically classify the homes of organisms into roughly ten different regional climate communities called **biomes**. **Biomes are described in terms of abiotic factors such as climate and soil type, and biotic factors such as plant and animal life.** Each biome is associated with seasonal patterns of temperature and precipitation that can be summarized in a graph called a climate diagram, like the one in **Figure 3-15**. The distribution of major biomes is shown in **Figure 3-16**. Note that even within a defined biome, there is often considerable variation among plant and animal communities. These variations can be caused by differences in exposure, elevation, or local soil conditions. Local conditions also can change over time.

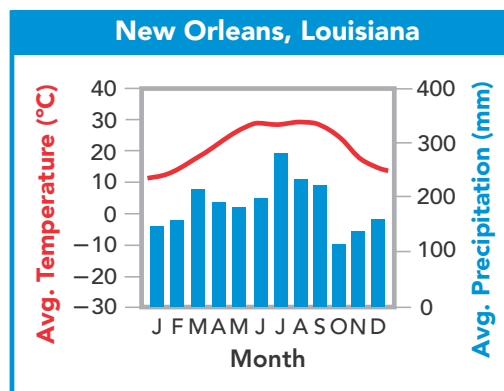


Figure 3-15
Climate Diagram

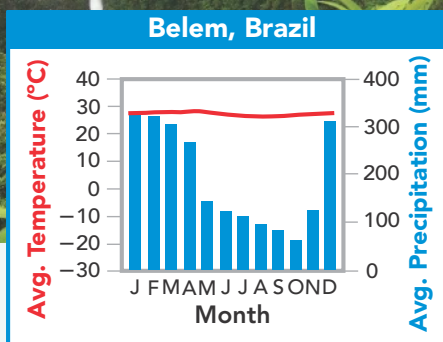
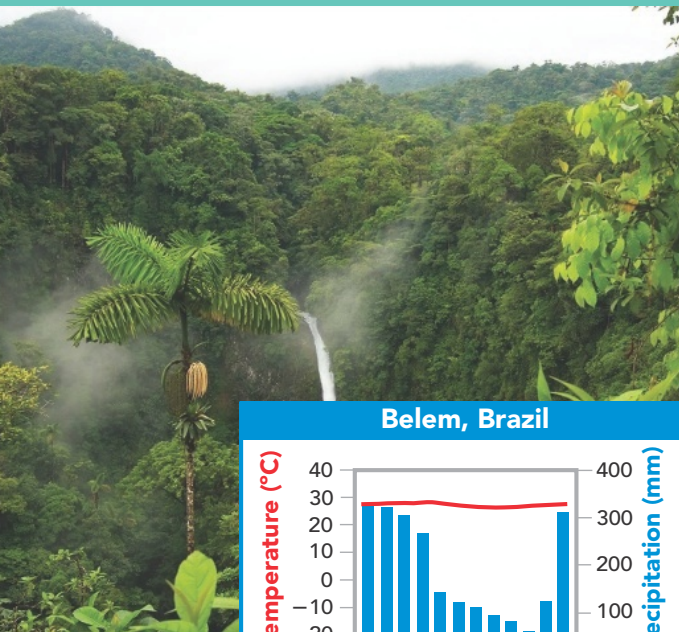
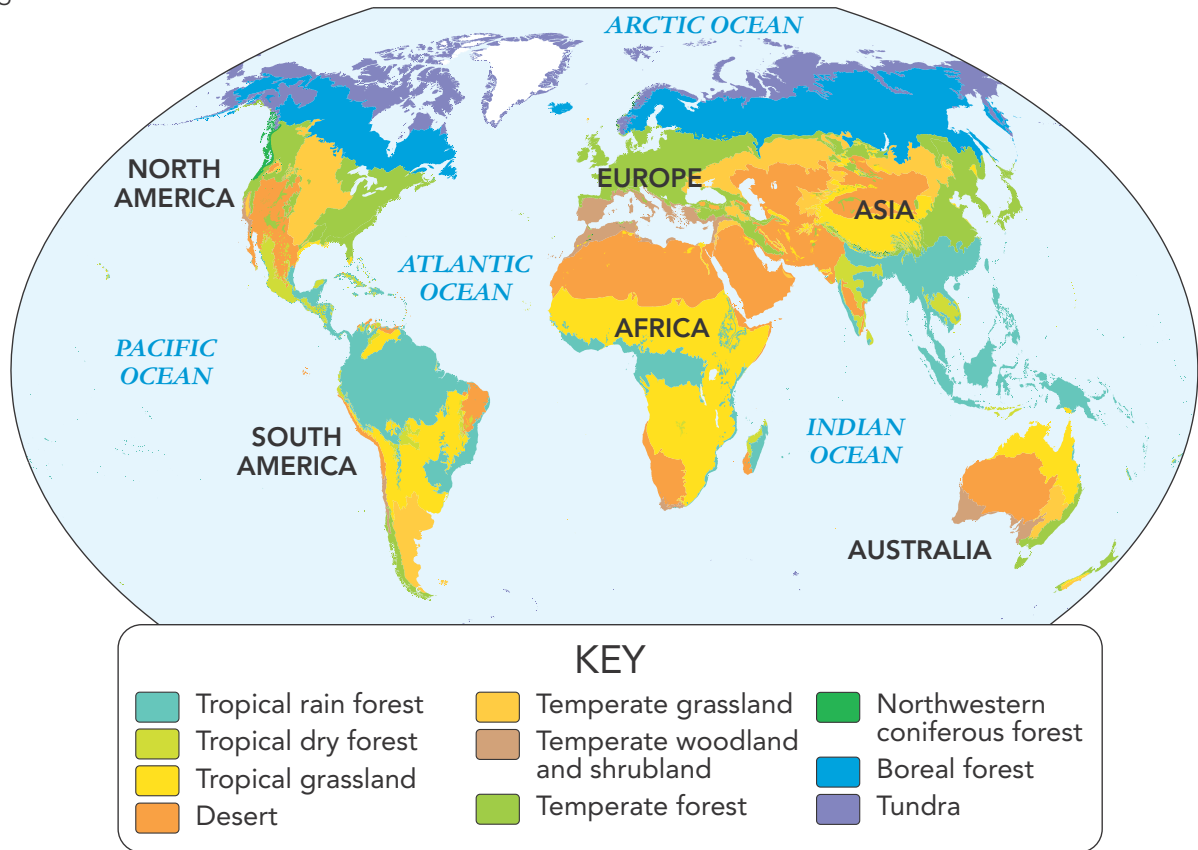
A climate diagram shows the average temperature and precipitation of a given location during each month of the year. In this graph, and those to follow, temperature is plotted as a red line, and precipitation is shown as vertical blue bars.

Visual Summary

Figure 3-16

Biomes

This map shows the locations of the world's major biomes. Each biome has a characteristic climate and community of organisms.



TROPICAL RAIN FOREST

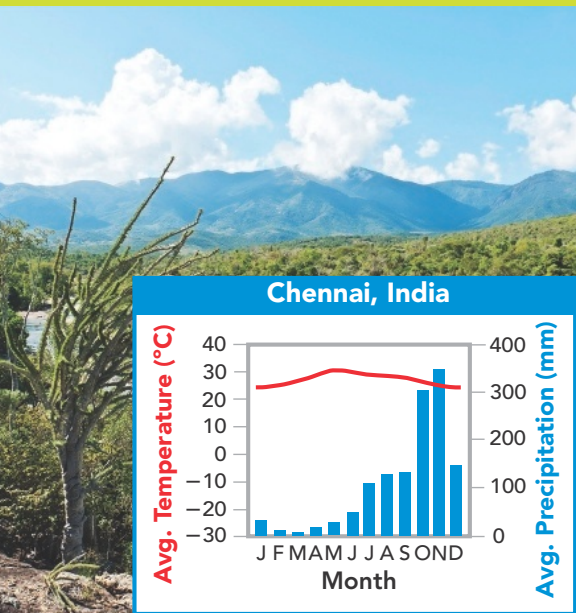
Tropical rain forests have more species than all of the other biomes combined. Tropical rain forests typically get at least 2 meters of rain annually. Tall trees form a dense, leafy covering called a **canopy** from 50 to 80 meters above the forest floor. In the shade below, shorter trees and vines form a layer called the **understory**. Organic matter on the forest floor is recycled and reused so quickly that the soil in most tropical rain forests is not very rich in nutrients.

Abiotic factors: warm and wet year-round; thin, nutrient-poor soils subject to erosion

Biotic factors

Plant life: Understory plants compete for sunlight, so many have large leaves to capture light. Tall trees growing in shallow soil often have buttress roots, which act like props for support. Epiphytic plants grow on the branches of tall plants, taking advantage of available sunlight.

Animal life: Animals are active all year. Many use camouflage to hide from predators; some can change color to match their surroundings. Animals in the canopy have adaptations for climbing, jumping, and flight.



TROPICAL DRY FOREST

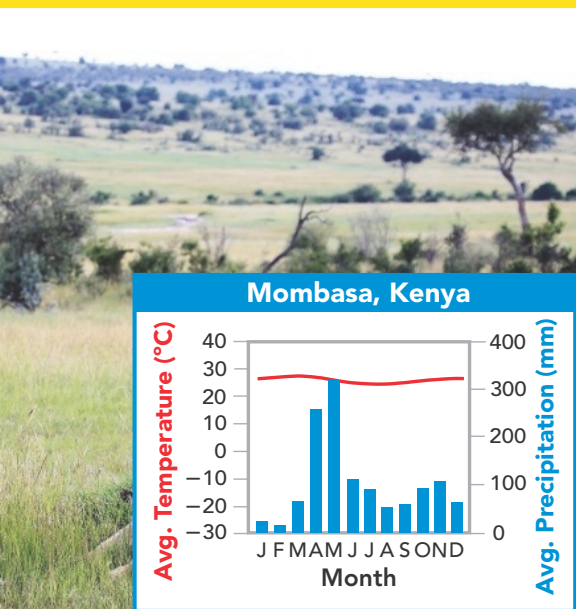
Tropical dry forests grow where rainy seasons alternate with dry seasons. In most places, a period of rain is followed by a prolonged period of drought.

Abiotic factors: warm year-round; alternating wet and dry seasons; rich soils subject to erosion

Biotic factors

Plant life: Many plants here lose their leaves during the dry season. Some leaves have an extra thick waxy layer to reduce water loss. Others store water in their tissues.

Animal life: Many animals reduce their need for water by entering long periods of inactivity similar to hibernation, but typically taking place during a dry season. Other animals, including many birds and primates, move to areas where water is available during the dry season.



TROPICAL GRASSLAND/SAVANNA/SHRUBLAND

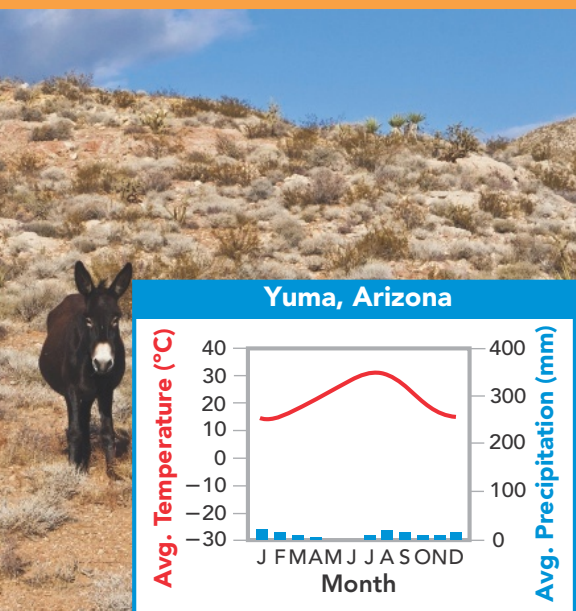
This biome receives more seasonal rainfall than deserts but less than tropical dry forests. Grassy areas are spotted with isolated trees and small groves of trees and shrubs.

Abiotic factors: warm; seasonal rainfall; compact soils; frequent fires set by lightning

Biotic factors

Plant life: Plant adaptations are similar to those in the tropical dry forest, including waxy leaf coverings and seasonal leaf loss. Some grasses have a high silica content that makes them less appetizing to grazing herbivores. Also, grasses have adaptations that enable them to continue to grow after being grazed.

Animal life: Many animals migrate in search of water during the dry season, while some smaller animals burrow and remain dormant.



DESERT

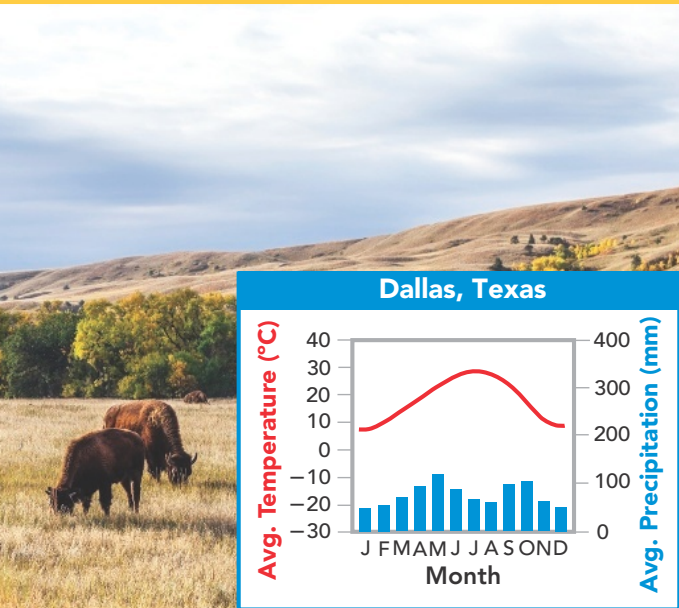
Deserts receive less than 25 centimeters of precipitation annually, but vary greatly depending on elevation and latitude. Many deserts undergo extreme daily temperature changes between hot and cold.

Abiotic factors: low precipitation; variable temperatures; soils rich in minerals but poor in organic material

Biotic factors

Plant life: Many plants, including cacti, store water in their tissues, and minimize leaf surface area to cut down on water loss. Cactus spines are actually modified leaves.

Animal life: Many desert animals get the water they need from their food. To avoid the heat of the day, many are active only at night. Blood vessels near the skin help animals lose body heat.



TEMPERATE GRASSLAND

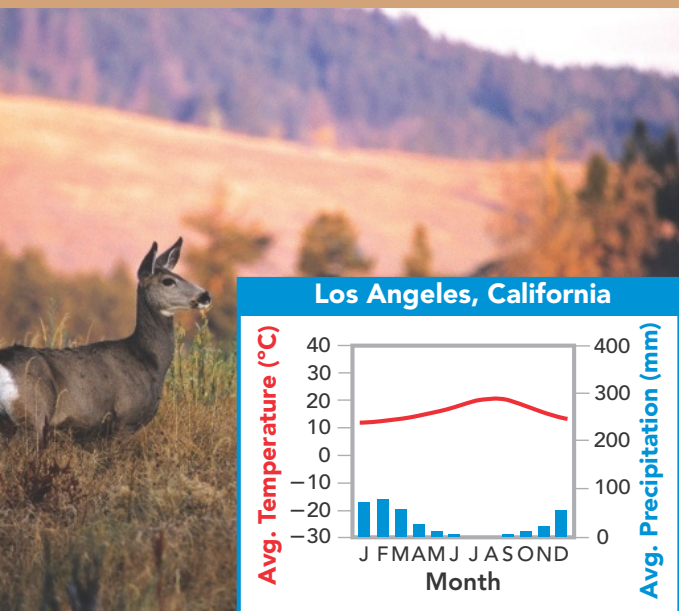
Plains and prairies with fertile soils once covered much of the Midwestern and central United States, land that is now used for farms. Periodic fires and heavy grazing by herbivores maintained the plant communities dominated by grasses.

Abiotic factors: warm to hot summers; cold winters; moderately seasonal precipitation; fertile soils; occasional fires

Biotic factors

Plant life: Grassland plants, especially grasses, are resistant to grazing and fire. Dispersal of seeds by wind is common. The root structure of native grassland plants helps establish and retain deep, rich, fertile topsoil.

Animal life: These open, exposed environments make predation a constant threat for smaller animals. Camouflage and burrowing are common protective adaptations.



TEMPERATE WOODLAND AND SHRUBLAND

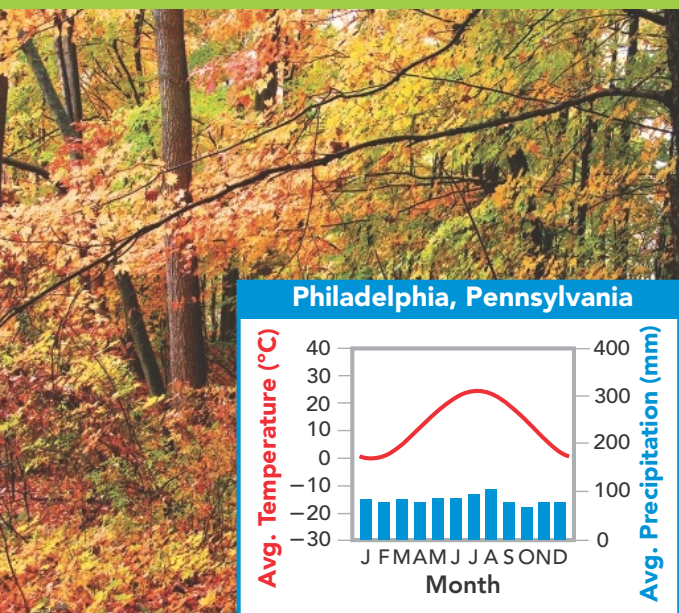
Here, large areas of grasses and wildflowers such as poppies are interspersed with oak and other trees. Communities that are more shrubland than forest are known as chaparral. Dense, low plants that contain flammable oils make fire a constant threat.

Abiotic factors: warm, dry summers; cool, moist winters; thin, nutrient-poor soils; periodic fires

Biotic factors

Plant life: Woody chaparral plants have tough waxy leaves that resist water loss. Fire resistance is important, although the seeds of some plants need fire to germinate.

Animal life: Animals tend to eat varied diets of grasses, leaves, shrubs, and other vegetation. In exposed shrubland, camouflage is common.



TEMPERATE FOREST

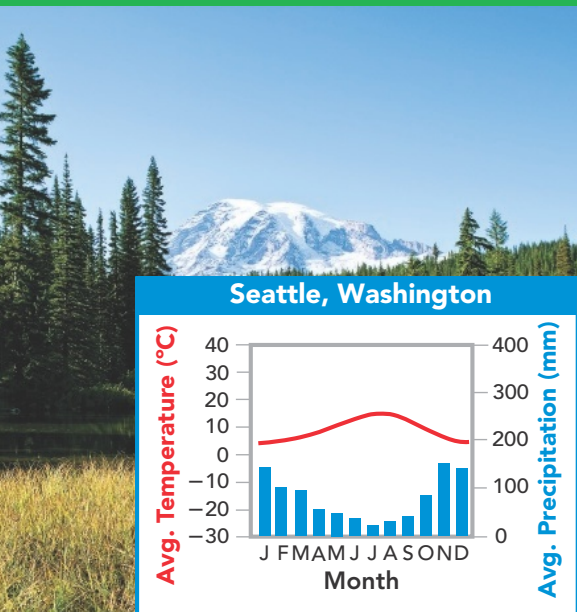
Temperate forests have cold winters and warm summers. Coniferous trees produce seed-bearing cones. In autumn, deciduous trees shed their leaves. Fertile soils are often rich in **humus**, a material formed from decaying leaves and other organic matter.

Abiotic factors: cold to moderate winters; warm summers; year-round precipitation; fertile soils

Biotic factors

Plant life: Deciduous trees drop their leaves and go into a state of dormancy in winter. Conifers have needlelike leaves that minimize water loss in dry winter air.

Animal life: Animals may cope with changing weather by hibernating or migrating to warmer climates. Other animals may be camouflaged to escape predation in the winter when bare trees leave them more exposed.



NORTHWESTERN CONIFEROUS FOREST

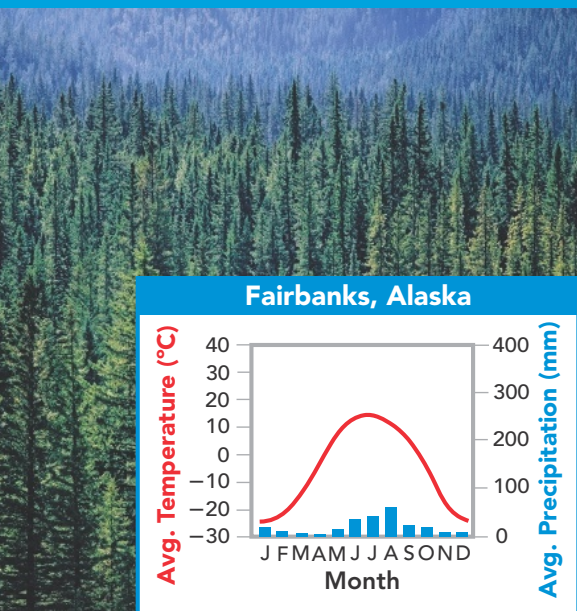
Mild, moist air and abundant rainfall nurture many tall conifers, from giant redwoods to spruce, fir, and hemlock, along with flowering trees and shrubs. Moss often covers tree trunks and the forest floor. This biome is sometimes called a “temperate rain forest.”

Abiotic factors: mild temperatures; abundant precipitation in fall, winter, and spring; cool, dry summers; rocky, acidic soils

Biotic factors

Plant life: Seasonal temperature variation means that there is less diversity here than in tropical rain forests. However, ample water and nutrients support lush, dense plant growth.

Animal life: Camouflage helps insects and ground-dwelling mammals avoid predation. Many animals are browsers—they eat a varied diet—an advantage in an environment where vegetation changes seasonally.



BOREAL FOREST/TAIGA

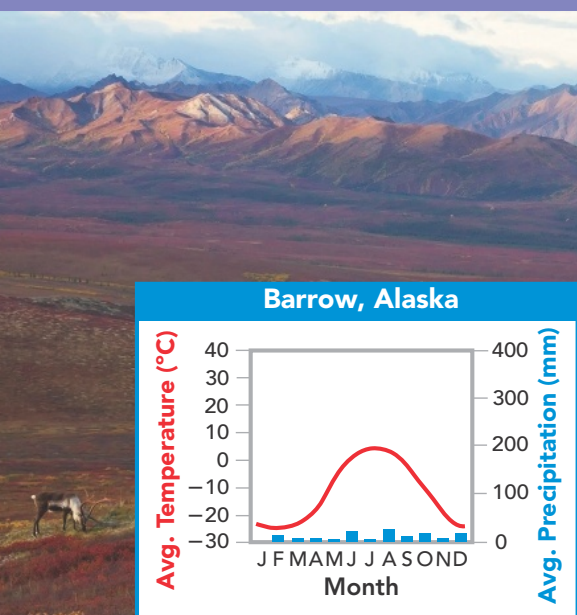
The word *boreal* comes from the Greek word for “north.” Dense forests of coniferous evergreens along the northern edge of the temperate zone are called boreal forests, or **taiga** (ty guh). Winters are bitterly cold, but summers are mild and long enough to allow the ground to thaw.

Abiotic factors: long, cold winters; short, mild summers; moderate precipitation; high humidity; acidic, nutrient-poor soils

Biotic factors

Plant life: Conifers’ shape and wax-covered needlelike leaves shed snow, and prevent excess water loss.

Animal life: Staying warm is the major challenge. Most have small extremities and extra insulation in the form of fat or downy feathers. Some migrate to warmer areas in winter.



TUNDRA

The tundra is characterized by **permafrost**, a layer of permanently frozen subsoil. During the short, cool summer, the ground thaws to a depth of a few centimeters and becomes soggy. This cycle of thawing and freezing rips and crushes plant roots, which is one reason that tundra plants are small and stunted. Cold temperatures, high winds, a short growing season, and humus-poor soils also limit plant height.

Abiotic factors: strong winds; low precipitation; short and soggy summers; long, cold, dark winters; thin soils

Biotic factors

Plant life: Ground-hugging mosses and other low-growing plants avoid damage from frequent strong winds. Seed dispersal by wind is common.

Animal life: Many animals migrate to avoid winters. Year-round residents display adaptations, among them natural antifreeze, small extremities that limit heat loss, and a varied diet.

Polar Regions Polar regions border the tundra and are cold year-round. Plants are few, though some algae grow on snow and ice. Where rocks and ground are exposed seasonally, mosses and lichens may grow. Marine mammals, insects, and mites are the typical animals. In the north, where polar bears live, the Arctic Ocean is covered with sea ice, although more and more ice is melting each summer. In the south, the continent of Antarctica, inhabited by many species of penguins, is covered by ice nearly 5 kilometers thick in places.



Biomes Today There's just one problem with the traditional concept of biomes: There aren't many natural communities left. In our modern world, humans have altered nearly 75 percent of all land outside the steepest mountain slopes, the coldest polar regions, and the driest deserts. If you go outside your school or home, the overwhelming odds are that there won't be one shred of an original, untouched biome anywhere near you.

The concept of biomes, however, remains useful for describing large regions in which natural biological systems contain certain organisms and function in certain ways. Those regions differ from one another in terms of climate and other environmental factors in ways that are useful to understand, even if the original ecosystem is no longer there. And, as you will learn later in this unit, areas that have been modified by humans are still influenced by the same factors that shaped the ecosystems before our species came along.



INTERACTIVITY

Study three organisms to determine what biomes they are best suited for.

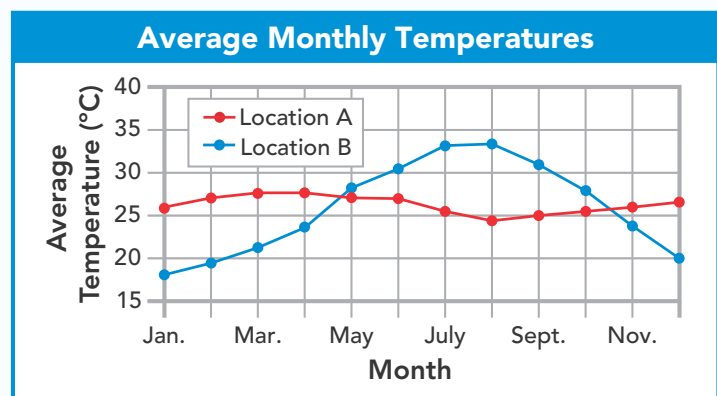
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Analyzing Data

Which Biome?

An ecologist collected climate data from two locations. The graph shows the monthly average temperatures in the two locations. In Location A, the total yearly precipitation is 273 cm. In Location B, the total yearly precipitation is 11 cm.

- Analyze Graphs** What specific question is this graph addressing?
- Analyze Graphs** Use the graph as evidence to draw a conclusion about the temperature over the course of the year in Location A and Location B.
- Apply Scientific Reasoning** In which biome would you expect to find each location, given the precipitation and temperature data? Use scientific reasoning to explain your answer.
- Construct Graphs** Look up the average monthly temperature last year for your community. Construct a graph and plot the data. Then, research the monthly rainfall for your city, and plot those data on your graph. Based on your results, which biome do you live in? Did the data predict the biome correctly?



Marine Ecosystems

Similar to organisms living on land, underwater organisms are affected by external environmental factors. **Aquatic ecosystems are described primarily by salinity, depth, temperature, flow rate, and concentrations of dissolved nutrients.** There are three main groups of aquatic ecosystems: marine ecosystems, freshwater ecosystems, and estuaries.

Just as biomes typically occupy certain latitudes and longitudes, marine ecosystems typically occupy specific areas within the ocean. Ecologists usually divide the ocean into zones based on depth and distance from shore, as shown in **Figure 3-17**.

Water depth influences aquatic life because sunlight doesn't penetrate far in water. The sunlit region near the surface in which photosynthesis can occur is known as the **photic zone**. The photic zone may be as deep as 200 meters in tropical seas, but just a few meters deep or less in rivers and swamps. Below the photic zone is the dark **aphotic zone**, where photosynthesis cannot occur.

Food chains in many aquatic ecosystems are based on **plankton**, a term that includes floating algae—called phytoplankton—and small, free-swimming animals called zooplankton. Phytoplankton need enough light for photosynthesis, so they can grow only in the photic zone. Zooplankton, many of which feed on phytoplankton, may swim up and down, in and out of the photic zone.

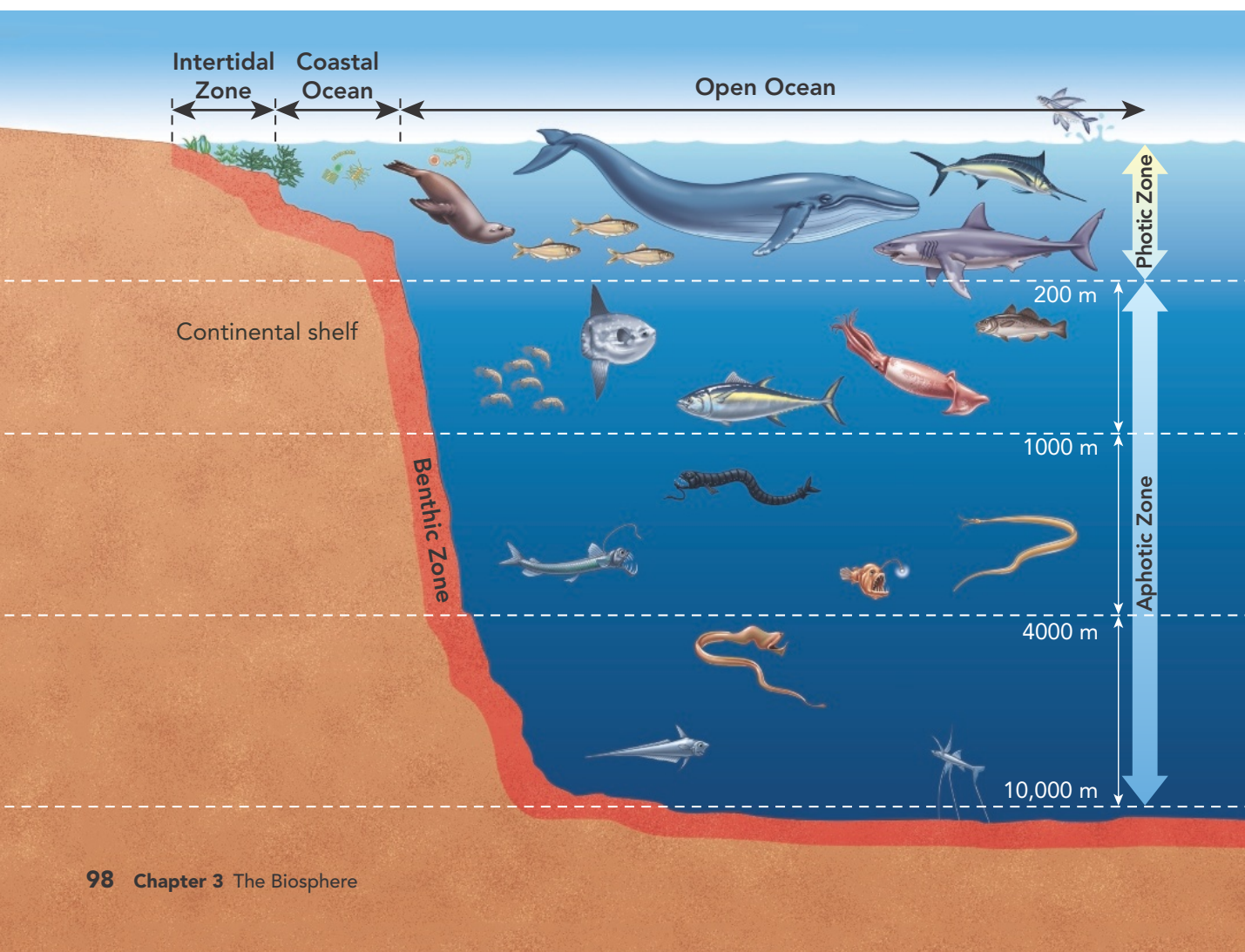
BUILD VOCABULARY

Prefixes The prefix *photo-* refers to light. A photic zone is well-lit, and photosynthesis uses the energy of light.

Figure 3-17

Ocean Zones

The ocean can be divided vertically into zones based on light penetration and depth, and horizontally into zones based on distance from shore.



Intertidal Zone Organisms in the intertidal zone are submerged in seawater at high tide and exposed to air and sunlight at low tide. These organisms experience regular and extreme changes in temperature, and are often battered by waves and currents. There are many different types of intertidal communities. A typical rocky intertidal community exists in temperate regions where exposed rocks line the shore. There, barnacles and seaweed permanently attach themselves to the rocks.

Coastal Ocean The coastal ocean extends from the low-tide mark to the outer edge of the continental shelf—the relatively shallow border that surrounds the continents. Water here is brightly lit, and is often supplied with nutrients by freshwater runoff from land. As a result, coastal oceans tend to be highly productive. Kelp forests and coral reefs are two important coastal communities. A coral reef is shown in **Figure 3-18**.

Open Ocean The open ocean begins at the edge of the continental shelf and extends outward. More than 90 percent of the world's ocean area is open ocean. Depth ranges from about 500 meters along continental slopes to more than 10,000 meters in deep ocean trenches. The open ocean can be divided into two main zones according to light penetration: the photic zone and the aphotic zone.

Open Ocean Photic Zone The open ocean typically has low nutrient levels and supports only the smallest species of phytoplankton. Still, the sunlit top 100 meters of the open ocean cover much of Earth's surface. So, most photosynthesis occurs here—and not in rain forests or other terrestrial environments.

Open Ocean Aphotic Zone The permanently dark aphotic zone includes the deepest parts of the ocean, where no photosynthesis can occur. Deep ocean organisms are exposed to high pressure, frigid temperatures, and total darkness. The deep ocean floor was once thought to be nearly devoid of life but is now known to have islands of high productivity. Deep-sea vents, where superheated water boils out of cracks on the ocean floor, support entire ecosystems based on chemical energy.

 **READING CHECK Compare** How does the coastal ocean zone compare with the open ocean zone?



Figure 3-18
Coastal Ocean:
Coral Reef

Tropical coral reefs are among the most diverse and productive communities on Earth.



INTERACTIVITY

Learn about how factors such as chemistry, light, salinity, and temperature are influenced at different depths.

Freshwater Ecosystems

Only 3 percent of Earth's surface water is fresh water, but that small percentage provides terrestrial organisms with drinking water, food, and transportation. Often, a chain of streams, lakes, and rivers begins in the interior of a continent and flows through several biomes to the sea.

Q Freshwater ecosystems can be divided into three main categories: rivers and streams, lakes and ponds, and freshwater wetlands.



RIVERS AND STREAMS

Rivers, streams, creeks, and brooks often originate from underground water sources in mountains or hills. Near a source, water has plenty of dissolved oxygen but little plant life. Downstream, sediments build up and plants establish themselves. Still farther downstream, water may meander slowly through flat areas. Animals in many rivers and streams depend on terrestrial plants and animals that live along their banks. Often, a chain of streams and rivers begins in the interior of a continent and flows through several biomes to the sea.

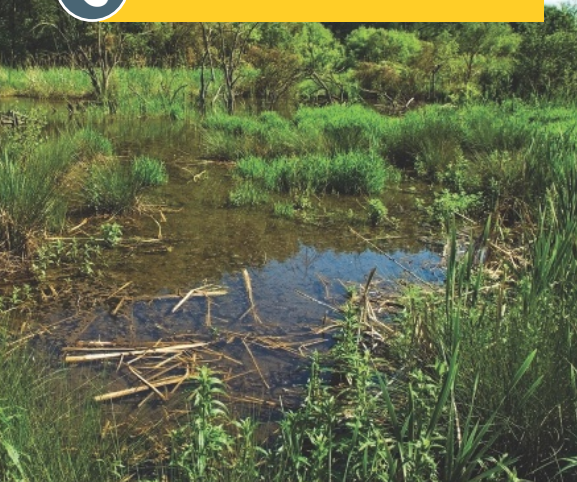


LAKES AND PONDS

Much of the life in lakes and ponds depends on a combination of plankton and attached algae and plants. Water typically flows in and out of lakes and ponds, often through rivers or streams. Water also circulates between the surface and the bottom during at least some seasons. This circulation distributes heat, oxygen, and nutrients. During a cold winter, thick ice might cover the lake for many months. Fish and other animals still live in the liquid water beneath.



INTERACTIVITY



FRESHWATER WETLANDS

A **wetland** is an ecosystem in which water either covers the soil or is present at or near the surface for at least part of the year. Water may flow through freshwater wetlands or stay in place. Wetlands are often nutrient-rich and highly productive, and they serve as breeding grounds for many organisms. Freshwater wetlands have important environmental functions: they purify water by filtering pollutants and help to prevent flooding by absorbing large amounts of water and releasing it slowly. Three main types of freshwater wetlands are freshwater bogs, freshwater marshes, and freshwater swamps.

Estuaries

An **estuary** (ES tyoo er ee) is a wetland formed where a river meets the sea. Fresh water and salt water often mix here, rising and falling with ocean tides. **Estuaries serve as spawning and nursery grounds for many ecologically and commercially important fish and shellfish.** The species include bluefish, striped bass, shrimp, and crabs.

Salt marshes are temperate estuaries characterized by salt-tolerant grasses above the low-tide line and seagrasses below water. One of the largest salt marshes in the United States surrounds the Chesapeake Bay in Maryland.

Mangrove swamps are tropical estuaries characterized by several species of salt-tolerant trees, collectively called mangroves. The largest mangrove area in America is part of Florida's Everglades National Park, shown in **Figure 3-19**.



Figure 3-19
Mangrove Swamps

Mangroves have several adaptations for growing in salty water. Roots that grow down from branches, called prop roots, help stabilize the plant in the damp soil.

HS-LS2-2

LESSON 3.3 Review

KEY QUESTIONS

1. What abiotic and biotic factors describe a biome? Give examples for a specific biome.
2. What factors define aquatic ecosystems?
3. Why are wetlands important?
4. Where are estuaries found?

CRITICAL THINKING

5. **Connect to Society** Many years ago, a wetland area was filled in to build a baseball field and playground. Now the town proposes to restore

the area to become a shallow pond surrounded by a marsh. What do you think are the strongest arguments for and against this plan? What information would help people decide whether or not to support the plan?

6. **Critique** Shrublands are often referred to as *scrub* or *scrubland*. Why do you think the authors decided to use the term *shrubland* instead? Do you agree or disagree with their choice?